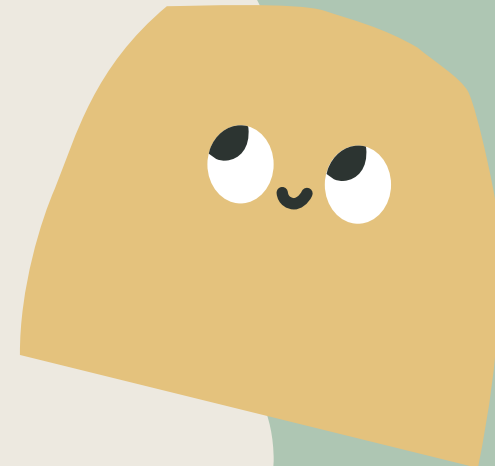
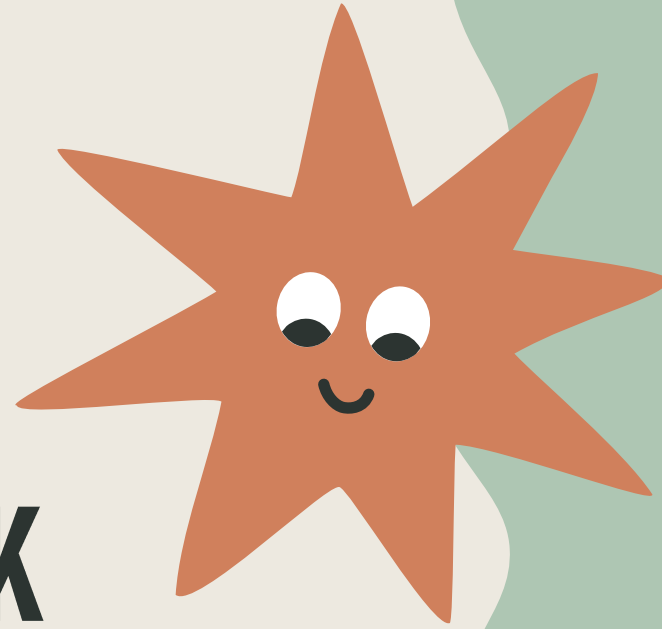


UNIT 4

DATA COMMUNICATION AND COMPUTER NETWORK

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

AWINASH GOSWAMI



Agenda

Basics of Data Communication

Basic Terminologies of Data Communication

Components of a communication system

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Transmission medium

Transmission impairments

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Basics of computer networks

Types of computer networks

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Basics of Data Communication

- Communication is the process of sharing a message. A conversation between two people is an example of communication.
- Data communications refers to the sharing of a virtual message between two devices. Electronic communications, like emails and instant messages and phone calls are examples of data communications.

Basic Terminologies of Data Communication

Data

The raw facts and figures are called data. The word data is derived from Latin language and it is plural of Datum. The text, numbers, symbols, images, voice and video which are also data. Data can be considered as unprocessed information.

Data Communication

Data Communication is the process of transferring data electrically from one place to another.

Data Transmission

The data transmission means emission of data in any direction via wireless or wired medium.

Analog Signals

Analog signals are waves that change with time period and used to represent data. An analog signal can be used to measure changes in some physical quantities such as light, sound, pressure or temperature.

Basic Terminologies of Data Communication

Digital Signals

A digital signal is an electrical signal represented as 0 or 1.

Data Rate / Bit Rate

Data Rate = **speed of data transfer** in bits per second (bps/kbps/Mbps).
Example: A 4G connection with 20 Mbps downloads 20 million bits every second.

Signal to Noise Ratio

SNR (Signal-to-Noise Ratio) measures how strong the signal is compared to the noise. A higher SNR means clearer communication.

Components of a communication system

Message	It is the information or data to be communicated. Common forms of information include text, numbers, pictures, audio and video.
Sender	It is the device that generates and sends a message. It can be a computer, telephone handset, etc.
Receiver	It is the device that receives a message. It can be a computer, telephone handset, etc.
Medium	It is the channel or path through which the message is carried from sender to the receiver. Some examples include twisted-pair cable, coaxial cable, radio waves, etc.
Protocol	A protocol is an agreement/rules between two parties(communication devices). Sender and receiver follow same protocols to communicate with each other. Examples include; SMTP, HTTPS, FTP, UDP etc.

Properties of a Good Communication System

Delivery

Making sure that the data is delivered is the first fundamental characteristic of any communication network. The system must be able to deliver data in correct order to the correct destination.

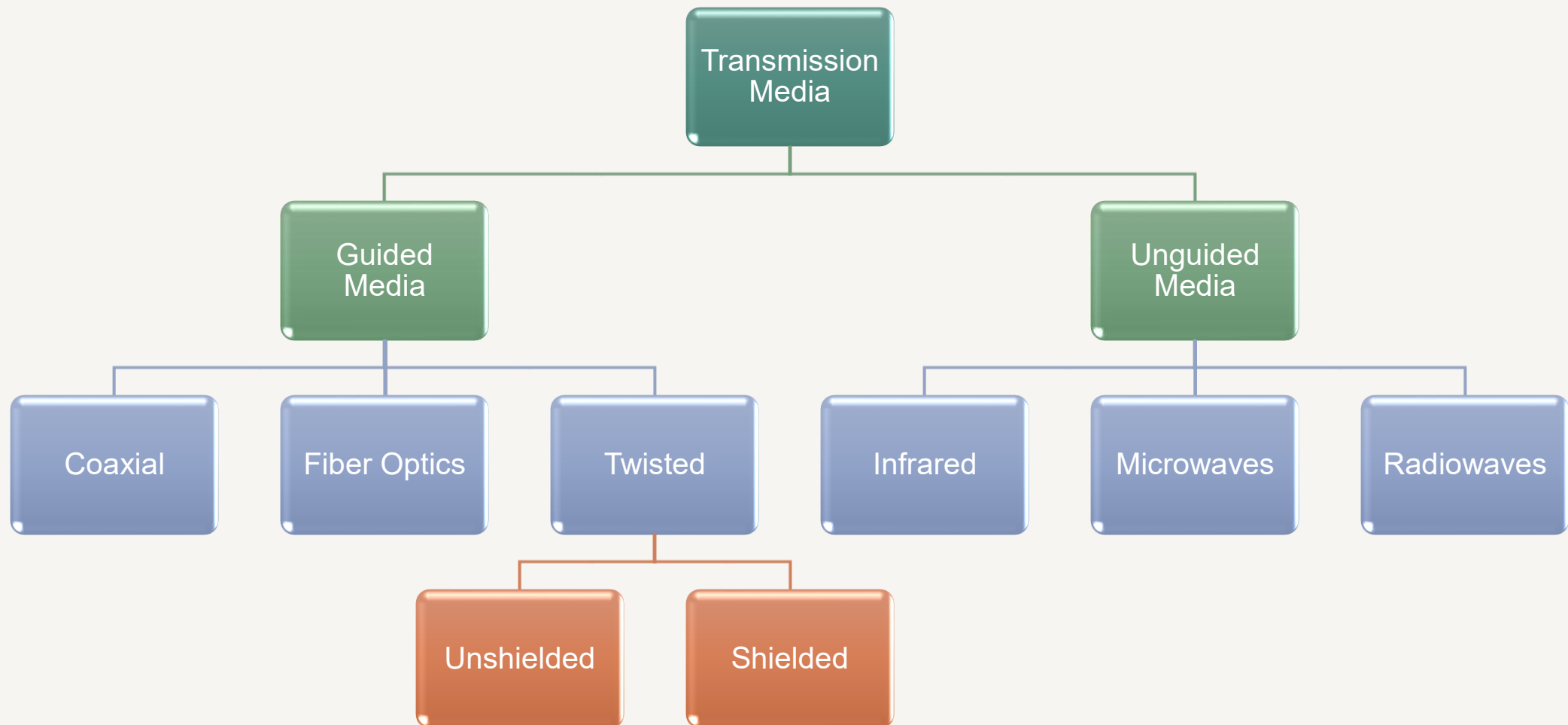
Accuracy

The system must deliver the data accurately. Data that has been altered during transmission and left uncorrected is not useful.

Timeliness

The data must be delivered in a timely manner. Late delivered data is useless.

Transmission Media



Guided Media

- In guided media, signals are transmitted using physical links (cables, wires). It is also called Wired or Bounded transmission media.

Guided Media: Twisted Pair Cable

- This cable is made by two separate wires twisted together. A twisted pair cable is made up of insulated copper wires. The insulation and twisting of wires prevent external interference. Each pair of wires has unique color code. This type of cable is widely used in different kinds of data and voice infrastructures. There are two types of twisted pair cables:

Guided Media: Twisted Pair Cable



Unshielded Twisted Pair (UTP)

This type of cable can block interference but it is vulnerable to external interference. It is mostly used for telephonic applications. It is less expensive and easy to install.



Shielded Twisted Pair (STP)

This type of cable consists of a special coating to block external interference. It is used in fast-data-rate ethernet and also in voice and data channels of telephone lines.

Guided Media: Coaxial Cable & Fiber-Optic Cable



Coaxial Cable

Coaxial cable is also known as coax. It has an outer plastic covering containing two parallel conductors each having a separate insulated protection cover. Cable TVs and analog television networks widely use coaxial cables



Fiber-Optic Cable

In optical fiber or fiber-optic cable data is transferred in the form of light. It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data at very high speed.

Unguided Media

- Unguided media is also termed as wireless or unbounded transmission media.
- It does not require physical medium such as wire for the transmission of signals.
- There are three major types of Unguided Media.

Unguided Media Cont....

Radio Waves

Radio waves = **omnidirectional electromagnetic waves** that penetrate buildings and need no antenna alignment — used in **radio, TV, cordless phones, Wi-Fi, Bluetooth, and mobile networks**.

Microwaves

Microwave transmission is a **line-of-sight** transmission, meaning sending and receiving antennas must be **properly aligned**. It is mainly used for **mobile phone towers and television broadcasts**. There are two types:

Terrestrial – Both antennas are on **Earth**.

Satellite – One antenna is on a **satellite in orbit**, others on Earth stations. Works well in **remote places** and for **mobile devices**.

Infrared

Infrared = light-based transmission (LED → photodiode) at terahertz frequency. Cannot pass through walls — line-of-sight only. Used for short-range links and wireless LANs. *Examples:* TV remote, wireless keyboards, some wireless LANs.

Transmission Impairments

- Sometimes, signals traveling through transmission media lose their quality. This means that received signal is not same as the signal that was sent. This phenomenon is called transmission impairments. There are three causes of impairment i.e. attenuation, distortion and noise.

Transmission Impairments

Attenuation

Attenuation = **signal weakening** over distance due to medium resistance. Fixed using **amplifiers**. Measured in **decibels (dB)**.
Example: Long cables need repeaters/amplifiers to boost the signal.

Distortion

Distortion = **change in signal shape** because different frequency components travel at different speeds and arrive at different times, causing **phase differences** between sender and receiver.
Example: A sharp pulse becoming blurred or spread out after traveling through a cable.

Noise

Noise is an **unwanted signal** that mixes with the original signal during data transmission and damages it. Common types include: **Induced noise** (from external sources like motors or appliances), **Crosstalk noise** – (interference from a nearby wire/cable).

COMMUNICATION DEVICES



Switch connects multiple devices on a network via cables so they can communicate with each other.

Example: In a school lab, all computers connect to one switch to share files and printers.



Router connects multiple networks, finds the best path for data, and controls network traffic.

Example: Your home Wi-Fi router connects your home network to the internet.



Modem: A modem (Modulator-Demodulator) is a device that connects your computer to the internet through a phone line, cable, or fiber connection. It modulates digital signals into analog signals for transmission and demodulates incoming analog signals back into digital data. *E.g.* Your home internet box that connects to the ISP.

COMMUNICATION DEVICES



Dial-up modem uses phone lines to send data by converting digital ↔ analog signals. *Example:* Old internet connection that used to make a dialing sound before connecting.



DSL Modem uses phone lines for high-speed digital data and voice. Has a built-in switch; some support Wi-Fi. E.g. Home broadband connection that lets you use internet and phone at the same time.



ISDN (Integrated Services Digital Network) modem digital phone connection that sends data, voice, and video together over phone lines. Faster but costly. E.g. Early video conferencing and high-speed internet before broadband became common.



NIC (Network Interface Card) hardware that connects a computer to a network by converting data into electrical signals. Usually built into modern motherboards. *Example:* The Ethernet port on your computer or laptop's Wi-Fi card.

BASICS OF COMPUTER NETWORKS

Computer Network

A computer network is a group of computers and related equipment connected to share data and other resources. The related equipment may be printer, scanners, fax machines, server, etc. The resources may include a file server, internet connection, etc.

Networking

Networking is the act of joining computers and its accessories so that exchange of information and sharing of resources take place.

Types of Computer Network

Local Area Network (LAN)

LAN = network of computers in a small area (home, school, office) to share internet, files, printers, and software using cheap hardware. E.g. Your home Wi-Fi network or a school computer lab.

Metropolitan Area Network (MAN)

MAN = network covering a city or campus (multiple LANs). Fast but expensive — used by large organizations and universities. Needs security to stop unauthorized access. E.g. A city-wide cable TV network or a university campus network connecting many buildings.

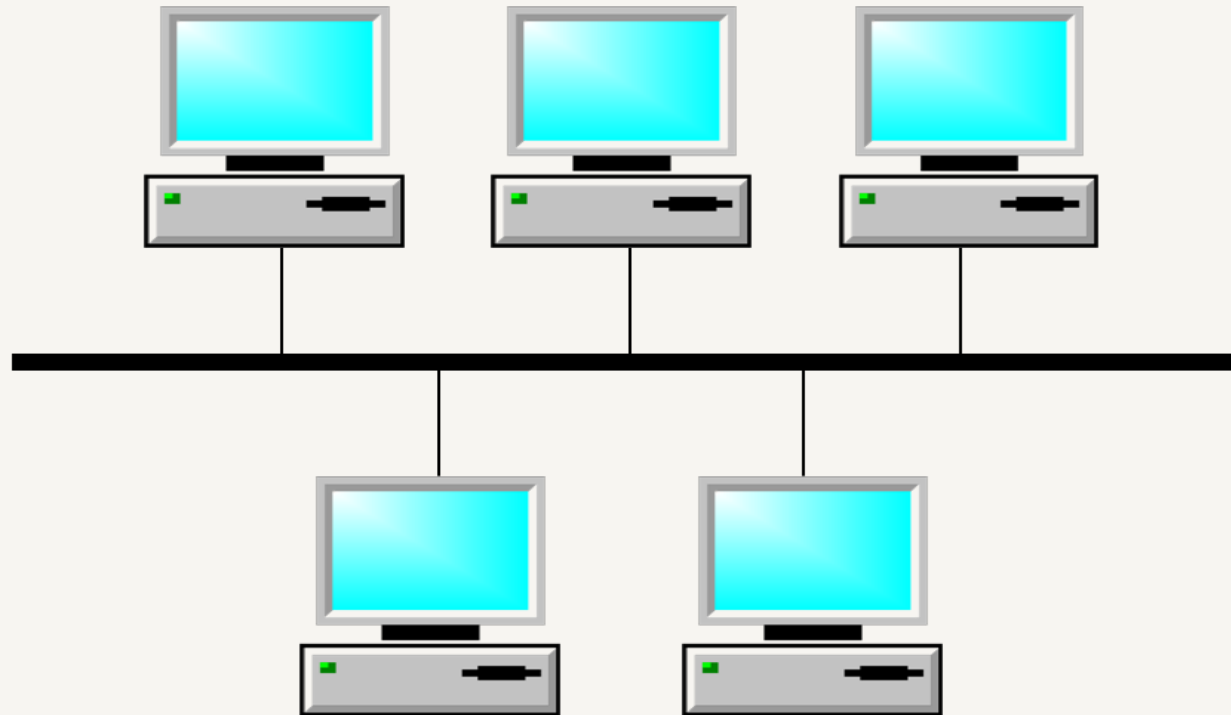
Wide Area Network (WAN)

WAN = network covering countries/continents (connects cities, MANs, LANs). Uses satellites, microwaves, telecom links. Expensive — used by large organizations. *E.g.* The internet itself is the largest WAN.

Fundamental of Topologies: Bus Topology

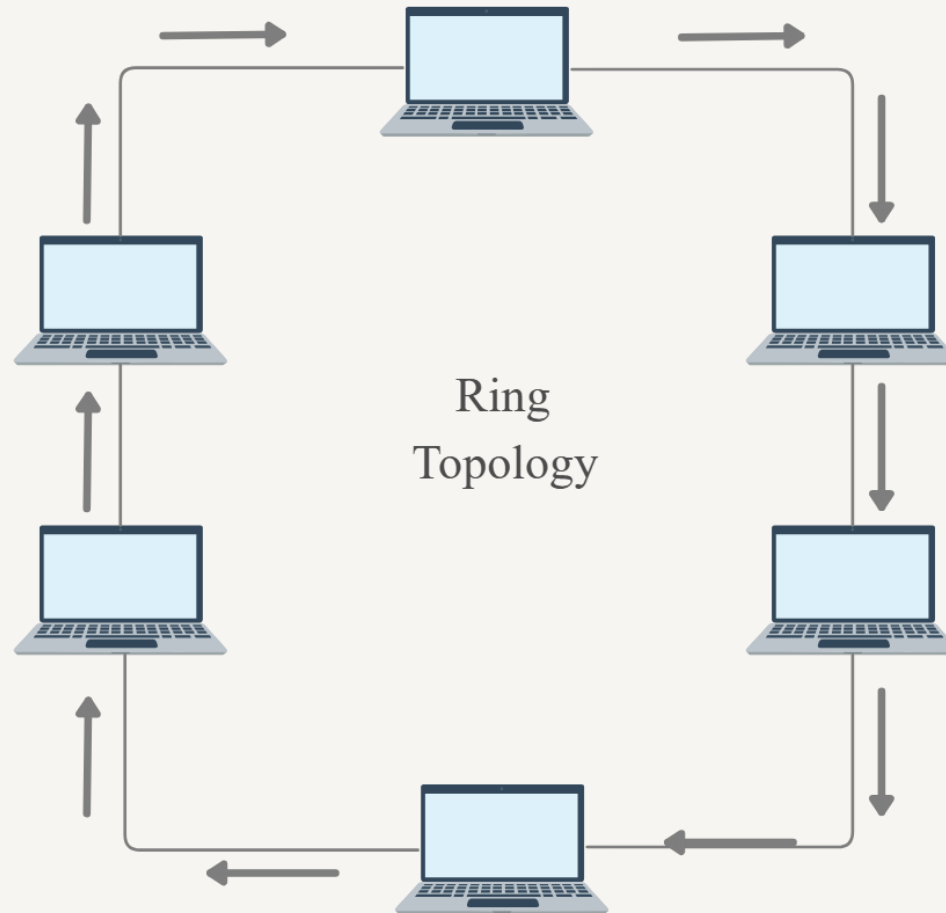
The physical layout in which computers are connected is called topology.

Bus Topology = **all devices on one cable (backbone)**. Cheap and simple, but **one cable failure kills the whole network**. *Example:* Old Ethernet networks using a single coaxial cable.



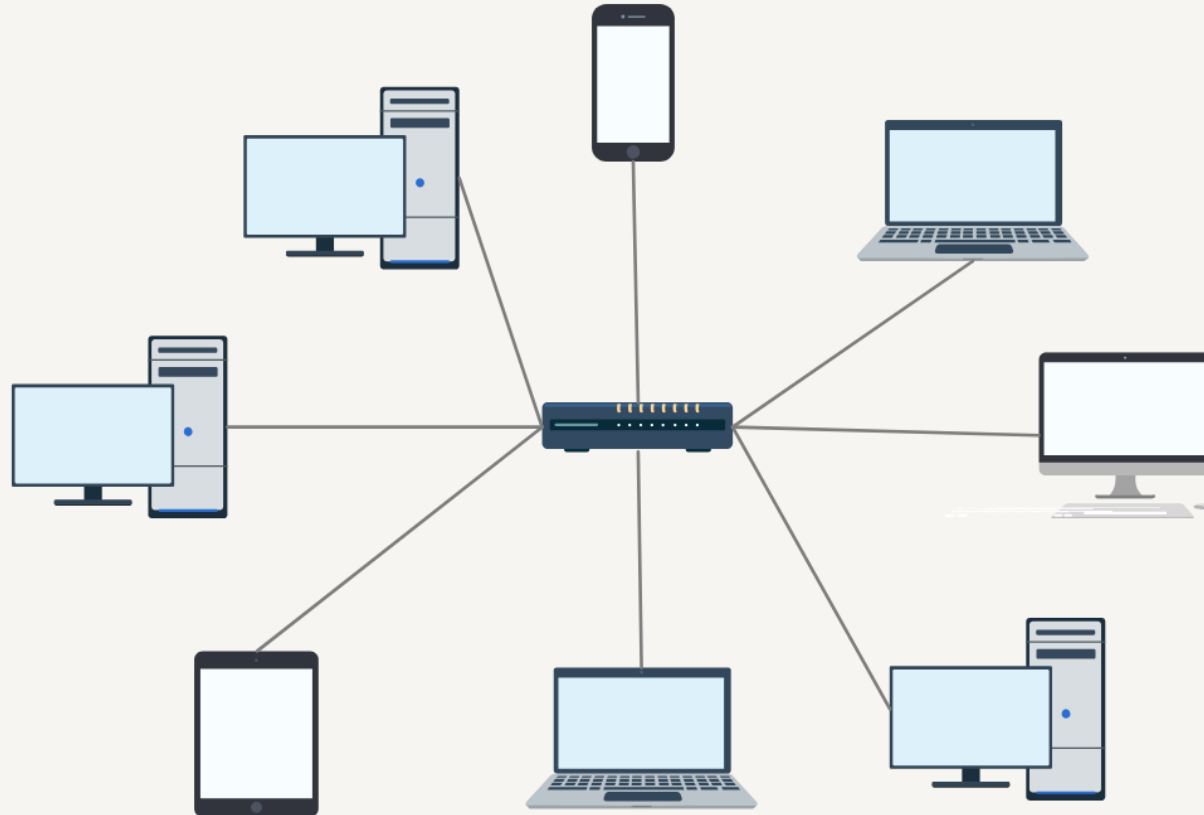
Fundamental of Topologies: Ring Topology

Ring Topology = **devices in a circle**, signal moves **one direction** through each device. **One failure can break the whole network**. *Example: Token Ring networks.*



Fundamental of Topologies: Star Topology

Star Topology = **all devices connect to a central hub/switch**. **Easy to set up and expand**; one broken link **doesn't crash the whole network**. *Example:* Most modern home and office networks use a central Wi-Fi router or switch.



NETWORK ARCHITECTURE: ISO's OSI Model

- **What is OSI Model?** Imagine you are **sending a letter to your friend in another city**.
The letter goes through many steps — you write it, put it in an envelope, write the address, post it, it travels by truck or plane, and finally reaches your friend.
- The **OSI Model** is exactly like that — but for **computer data**.
 - **OSI = Open Systems Interconnection**
 - It has **7 layers** (like 7 steps in sending a letter).
 - Each layer does a **specific job**.
 - Data passes **down** through the layers when sending and **up** through the layers when receiving.

NETWORK ARCHITECTURE: ISO's OSI Model

Layer	Name	Simple Job	Example
7	Application	What the user sees and uses	You open WhatsApp to send a message
6	Presentation	Translates, encrypts, compresses data	WhatsApp changes your text into a format the phone understands
5	Session	Starts, manages, and ends the connection	WhatsApp starts the chat session with your friend
4	Transport	Breaks data into packets, ensures no loss	Your message is split into small packets and numbered
3	Network	Finds the best path (routing)	Finds the best route from Karachi to Lahore
2	Data Link	Sends data between devices on the same network	Your phone sends data to the Wi-Fi router
1	Physical	Actual cables, signals, Wi-Fi waves	Wi-Fi signals or fiber optic cables carry the data

NETWORK ARCHITECTURE: ISO's OSI Model

Real-Life Example: Sending a WhatsApp Message

Let's say **Ali in Karachi** sends a WhatsApp message to **Ahmed in Hyderabad**.

- **Application (Layer 7):** Ali types "Salam" in WhatsApp.
- **Presentation (Layer 6):** The message is formatted, compressed, and encrypted.
- **Session (Layer 5):** A session is started between Ali's phone and Ahmed's phone.
- **Transport (Layer 4):** The message is broken into small **packets** with numbers (1, 2, 3...).
- **Network (Layer 3):** Each packet is given an **IP address** (like a house address) to find the best route.
- **Data Link (Layer 2):** Packets are given **MAC addresses** (like a CNIC number) for the local network.
- **Physical (Layer 1):** Data travels as **electrical signals or Wi-Fi waves** through cables or air.

On Ahmed's side, the process happens in **reverse order** — from Layer 1 up to Layer 7 — and Ahmed sees the message "Salam."

- **Mnemonic (Top to Bottom) - All People Seem To Need Data Processing | (Bottom to Top) Please Do Not Throw Sausages Pizza Away**

Why Do We Need the OSI Model?

- It helps **different companies' devices** work together.
- It makes **troubleshooting easier** — if WhatsApp isn't working, we can check which layer has the problem.
- It **standardizes** how data is sent across the world.

NETWORK ARCHITECTURE: TCP/IP Model

- **What is TCP/IP Model?** Imagine you are **sending a parcel through TCS or Leopards Courier** from Karachi to Sukkur. The parcel goes through a few main stages:
 - *You pack the item.*
 - *You write the address.*
 - *The courier company finds the best route.*
 - *The parcel travels by truck and reaches the destination.*
- The **TCP/IP Model** is exactly like that — but for **computer data on the internet**.
 - ***TCP/IP = Transmission Control Protocol / Internet Protocol***
 - *It is the **real model** used by the internet today.*
 - *It has **4 layers** (simpler than OSI's 7 layers).*
 - ***TCP/IP is what actually runs the internet — OSI is just a reference model.***

NETWORK ARCHITECTURE: TCP/IP Model

- **Real-Life Example: Sending a WhatsApp Message**
- Let's say **Ali in Karachi** sends a WhatsApp message to **Ahmed in Hyderabad**.
 - **Application (Layer 4):** Ali types "Salam" in WhatsApp. The app decides how to send it (using HTTP, SMTP, etc.).
 - **Transport (Layer 3):** The message is broken into small **packets** with numbers (1, 2, 3...). TCP makes sure all packets arrive safely.
 - **Internet (Layer 2):** Each packet is given an **IP address** (like a house address) to find the best route from Karachi to Hyderabad.
 - **Network Access (Layer 1):** Data travels as **electrical signals or Wi-Fi waves** through cables or air, using **MAC addresses** for local delivery.
- On Ahmed's side, the process happens in **reverse order** — from Layer 1 up to Layer 4 — and Ahmed sees the message "Salam."

NETWORK ARCHITECTURE: TCP/IP Model

- **Mnemonic - All Trains In Pakistan**

Layer	Name	Job	Example
4	Application	What the user sees and uses	You open WhatsApp or YouTube
3	Transport	Breaks data into packets, ensures delivery (TCP) or speed (UDP)	Your message is split into packets and numbered
2	Internet	Finds the best path using IP addresses	Finds the route from Karachi to Lahore
1	Network Access (Link)	Actual cables, Wi-Fi, signals, MAC addresses	Wi-Fi signals or fiber optic cables carry the data

TCP vs UDP (Simple Difference)

TCP	UDP
Reliable — all packets arrive	Fast — some packets may be lost
Used for: WhatsApp messages, emails, file downloads	Used for: Live streaming, video calls, online games
Like TCS courier — tracks every parcel	Like throwing a letter — fast but no tracking

Comparison: OSI vs TCP/IP

OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Application	Application
Presentation	Application
Session	Application
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	Network Access

- **Simple way to remember:**

- OSI's top 3 layers (Application, Presentation, Session) are **combined into one** in TCP/IP.
- OSI's bottom 2 layers (Data Link, Physical) are **combined into one** in TCP/IP.

Why Do We Need the TCP/IP Model?

- It is the **actual model used by the internet**.
- It helps **different devices** (phones, laptops, servers) communicate.
- It makes **troubleshooting easier** — if WhatsApp isn't working, we check which layer has the problem.
- It **standardizes** how data is sent across the world.

Difference between Physical Address and Logical Address

Physical Address	Logical Address
Physical address is attached with ROM of the NIC card.	Logical address is assigned to a device.
Physical Addressing means MAC (Media Access Control) provided by manufacturer and attached address of the NIC. The card which is used to connect your machine to the internet.	Logical addressing means IP addressing that is provided by your Internet Service Provider (ISP) or set by network administrator.
Physical addressing cannot be changed. They are also called hardware address.	Logical Address can be changed.
Physical address is a 48-bit MAC address.	Logical address is a 32-bit IP address.
It is globally unique and permanent.	It is unique in one network and temporary.

IPv4 Address

- Network address = **unique ID for a device on a network** (like a house address). Can be **physical** (MAC) or **logical** (IP).
Example: Your laptop has a unique IP address so data reaches only your device, not your neighbor's.
- An IP address is a unique number used to identify a device (computer, printer, smartphone, tablet, etc.) on a network. Every device connected to the internet must have an IP address to communicate with other devices. It works like a telephone number or car registration number — showing ownership and location.
- IPv4 (Internet Protocol version 4) addresses are made of 32 binary bits, divided into two parts:
 - Network portion – identifies the computer network
 - Host portion – identifies the specific device
- IPv4 addresses have four number segments separated by dots.
Example: 192.168.108.105